CHAPTER 13 · CONCEPT 1

Why Earth Is Unique: Its Position

Earth sits at just the right distance from the Sun for water to stay liquid — the key to life.

The right distance from the Sun

Of perhaps billions of planets, **Earth is the only one known to support life**. The most important reason is its **distance from the Sun —** just right for **liquid water** to exist.

- **Too close** → too hot, water evaporates.
- **Too far** → too cold, water freezes.
- The range of distances where water stays liquid is the **habitable zone** (or "Goldilocks zone").

> **Blue Planet:** about 70% of Earth's surface is water, so from space Earth looks blue.

The greenhouse effect

Gases like **carbon dioxide** trap some of the Sun's heat (after Earth is warmed), keeping the temperature **just right**. **Venus** is the **hottest planet —** not because it's closest to the Sun, but because its thick CO_2 atmosphere traps heat strongly (a runaway greenhouse effect).

> **Balance matters:** a mild greenhouse effect keeps Earth warm enough for liquid water; too much (like Venus) makes a planet far too hot.

Where students lose marks

> **Trap 1 — Venus is hottest, not Mercury.** Its thick CO_2 atmosphere traps heat, even though Mercury is closer to the Sun.

> **Trap 2 — The habitable zone is about LIQUID water.** Too hot → it evaporates; too cold → it freezes.

> **Trap 3 — The greenhouse effect is natural and useful in moderation.** It keeps Earth warm; only excess is harmful.

Model answers — write them like this

Example A. Why is Earth's distance from the Sun so important for life?

Step 1: It is just right — not too hot, not too cold.

Step 2: So water stays liquid, which is essential for life.

∴ The right distance keeps water liquid, allowing life.

Example B. Why is Venus hotter than Mercury, though Mercury is closer to the Sun?

Step 1: Venus has a thick atmosphere of carbon dioxide.

Step 2: It traps heat strongly (greenhouse effect).

∴ The trapped heat makes Venus the hottest planet.

Ready to practise? → Open the worksheet